

## Layer 3 — Research execution loops

### Product-owner view

**Product goal:** turn the prepared research workstation into real scientific execution loops.

Layer 1 prepared the runtime.

Layer 2 prepared the role workstations.

Layer 3 prepares the actual research machinery:

```
simulate
discover
train or test surrogate
map to symbolic DSL
search parameters
verify
run locally or on Runpod
```

This layer is where your actual MRes/research work starts becoming executable.

### Layer 3 bundles

Bundle 3 – NCA-ART research stack

Prepare the actual simulator/NCA/ART/DSL research execution loop.

Bundle 4 – Parameter search comparison tools

Prepare random/grid, Latin hypercube, evolutionary, Bayesian, and robustness workflows.

Bundle 5 – Runpod training/inference loop

Prepare the remote compute workflow for candidate batches, training, inference, checkpoints, and results

### Correct scientific direction

The research loop should follow this architecture:

1. Symbolic DSL / motif library  
Generate biologically constrained candidate GRNs.
2. PDE/ODE simulator  
Run reaction-diffusion simulations for 5-node GRNs.
3. Data capture  
Store per-cell state vectors, neighborhoods, timepoints, parameters, and metadata.
4. ART / ARTMAP discovery  
Cluster recurring local states into prototypes.  
Build transition graphs.  
Optionally learn neighborhood -> next-state mappings.
5. NCA  
Either train a fast surrogate from simulator trajectories,  
or test local update rules derived from ART transitions.
6. Prototype-to-DSL mapping  
Fit symbolic GRN parameters/topologies from promising prototypes and transitions.
7. Verification  
Run Turing/stability checks, parameter sweeps, robustness tests, and resimulation.

That matches the uploaded ART/NCA notes: the PDE/ODE simulator is the authoritative forward model; ART ingests simulator or NCA state vectors and produces prototypes and transition graphs; ARTMAP can help with input-to-next-state mappings; NCA can be a surrogate or alternate local update model; and DSL mapping turns prototypes into human-readable GRN candidates with constraints and verification.

## What Layer 3 must not do yet

not build the Agentfield controller  
not build the Paperclip adapter



```
not create dashboards
not automate full campaigns yet
not hide the scientific steps behind agents
```

Layer 3 should make the research loop work manually and reproducibly first.

## Layer 3 success condition

After Layer 3, you should be able to run a small local smoke experiment:

```
generate candidate 5-node GRNs
run a tiny PDE/ODE simulation
capture state snapshots
cluster with ART
produce a prototype/transition artifact
optionally train or test a tiny NCA surrogate
attempt prototype-to-DSL mapping
run one basic verification check
write outputs to /workspace/runs/nca-art-grn
```

Then Bundle 5 can move that same loop to Runpod.